MIL-E-1/1652A(NAVY)
2 October 1975
SUPERSEDING
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16 May 1968

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, VOLTAGE REGULATOR
1/ TYPE 8797

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Corona discharge

See figure 1

Mounting position: Any

Weight: 0.5 ounce (14.2 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ez	Eb	Ib	TA	Barometric pressure, reduced
Unit:	Vdc	Vdc	μAdc	°C	mmHg
Maximum:	950	930	550	+85	87
Minimum:	---	870	10	-55	---
TEST CONDITIONS:	---	---	100	25±5	---

GENERAL:

Qualification - Not required

Holding period (t = 7 days)

1/ Replaces Victoreen Type GV5B-900 and EIA Type 8797.

Ⓐ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
---	Holding period end points:								
3347	Ionization voltage	6		0.65	II	Ez	---	950	Vdc
---	Change in regulation voltage	6		0.65	II	ΔEb(1)	---	±5	Vdc
3335	Regulation	6,8		0.65	II	ΔEb	---	50	Vdc
---	Peak current	5,6		0.65	II	---	---	750	μAdc
3347	Ionization voltage	3		0.65	II	Ez	---	950	Vdc
3337	Voltage drop	-	Ebb/Ib = 100 ±2 μAdc	0.65	II	Eb(1)	878	922	Vdc
(A) 3335	Regulation	4	Ib = 10 to 550 μAdc	0.65	II	ΔEb	---	50	Vdc
---	Peak current	5		0.65	II	---	---	750	μAdc
	Quality conformance inspection, part 2								
3305	Leakage current	-	Eb = 850 Vdc (min); T = 25° ±5°C; Relative humidity = 85 ±5%	---	---	LIb	---	0.5	μAdc
1026	Low-temperature operation:								
3347	Ionization voltage	-	T = -55° ±5°C	---	---	Ez	---	950	Vdc
3337	Voltage drop	-	T = -55° ±5°C	---	---	Eb(1)	870	922	Vdc
3335	Regulation	8	T = -55° ±5°C	---	---	ΔEb	---	50	Vdc
1026	High-temperature operation:								
3347	Ionization voltage	-	T = 85° ±5°C	---	---	Ez	---	950	Vdc
3337	Voltage drop	-	T = 85° ±5°C	---	---	Eb(1)	878	930	Vdc
3335	Regulation	8	T = 85° ±5°C	---	---	ΔEb	---	55	Vdc
	Quality conformance inspection, part 3								
---	Life test	-	Group B; Ebb/Ib = 100 ±2 μAdc; t = 1,000 hours	---	---	---	---	---	---
---	Life-test end points:								
3347	Ionization voltage	-		---	---	Ez	---	950	Vdc
3337	Voltage drop	-		---	---	Eb(1)	870	930	Vdc
3335	Regulation	8		---	---	ΔEb	---	50	Vdc
(A) 1031	Variable-frequency vibration	1,7	No voltages applied; t = 5 minutes; 55 to 500 Hz; 10 G (max)	---	---	---	---	---	---
(A) 1042	Shock, specified pulse	2,7	Test condition C	---	---	---	---	---	---
(A) 1002	Barometric pressure, reduced	7	Pressure = 87 mmHg; Ebb = 950 Vdc	---	---	---	---	---	---

NOTES:

1. Criterion for passing this test shall be compliance after test with the initial requirements for voltage drop and regulation.
2. Electrical samples shall not be used for this test. Ten additional samples shall be submitted for shock test only. Criterion for passing this shall be compliance after test of at least 80 percent of the tubes with the initial requirements for voltage drop and regulation.
3. The tube under test must not have been conducting for at least 30 minutes prior to test. During this test the regulator tube shall not be exposed to external sources of radiation and shall be shielded from light. No conditioning current is permitted.
4. Eb maximum shall not be exceeded anywhere within the regulated current range. In testing for regulation, current shall be varied continuously through the regulated range.

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NOTES:

5. Peak current is the current maximum during a transient pulse which will not cause damage to the tube or cause tube to go into glow.

6. The holding period commences upon completion of the initial performance of the indicated tests. At the conclusion of the holding period the indicated tests will be performed again. Of the tubes failing the holding period end point test, those which fail with respect to regulated voltage only, may be retested after an additional 60-day period. $\Delta Eb(1)$ is the change in regulated voltage from beginning of holding period to end of holding period.

Ⓐ 7. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

8. "Regulation" is the difference between operating voltages at 10 and 550 μAdc.

User activities:
Navy - AS, OS, MC, CG, SH

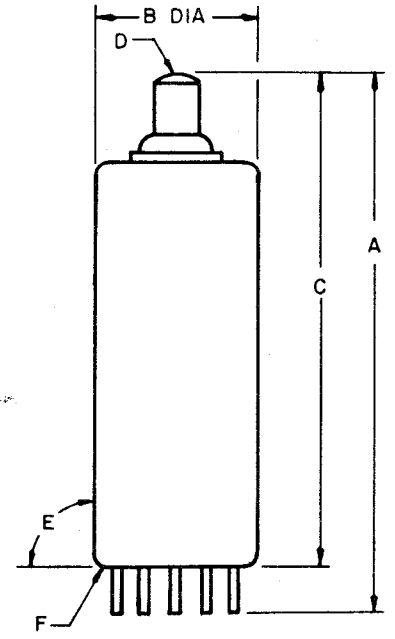
Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-N048)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		2.75 (44.45)
B		0.750 (19.05)
C		2.50
Ⓐ Quality conformance inspection, part 3 (see note)		
D	Cap: C1-3 (EIA)	
E	86.5°	93.5°
F	Base: E7-1 (EIA)	

PIN CONNECTIONS	
Pin No.	Element
1	k
2	k
3	k
4	k
5	k
6	k
7	k
cap	a

NOTE:
Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 8797.

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